MTP F26 · TEAM C · RADIO TEAM → ELECTRONICS TEAM

Suggested hardware for the transceiver

From Radio Team (Santiago, Andrian, Marcos) **To** Carlos, Michele, Bikash, Guillem

Version 0.4.3 · Fri 16 Oct 2026

New in v0.4.3 (16 Oct): low-battery countermeasures. The radio's 3.3 V LDO runs straight from the 3.0–4.2 V cell and that is fine (see Power budget); we added an ideal diode for battery sensing, firmware thresholds, a T0 check at 3.0 V and a buck-boost fallback.

New in v0.4.2 (16 Oct): a professor approved **20 dBm e.i.r.p.**, so SRI and MRM now run as a frequency-hopping radio at up to 20 dBm (+10.5 dB on the links). **No hardware change**: same SP4, same omni. The radio's TX bursts move toward the datasheet's **120 mA** in every SRI/MRM round, which the radio LDO, the 100 µF at the module and the battery path below already cover. Keep them sized for 120 mA.

New in v0.4 (decided 13 Oct): one Ebyte E01-ML01SP4 per box (nRF24L01+ with PA and LNA, surface-mount, antenna socket on the module) and **one flex omni antenna** plugged straight into it. No patch boards, no second radio. The SP4 is SMD, so it needs a small breakout from the first bench test.

The radio baseline is 2.4 GHz with the nRF24L01+. Under our split, the electronics team picks the micro and designs the board. This page is the radio team's **suggestion**, built around every hardware rule in the competition PDF, to save you time. It is not a decision. Please challenge it.

In short

1. **Micro module: Raspberry Pi Pico 2** (RP2350, **not** the Pico 2 W). It costs about $5, has 4 MB flash to hold the 1 MB file, 520 KB RAM and 3.3 V logic, and it can act as a USB host on any two pins or on its native USB port. It has no radio chip on board, so it is safe under the protocol ban.
2. **Radio: one Ebyte E01-ML01SP4 per box** (genuine nRF24L01+ with PA and LNA; SMD 14.85 × 18 mm, 10 pads at 1.27 mm; IPEX antenna socket). **VCC is 2.0–3.6 V only**, so it gets its own 3.3 V regulator. The same module goes from the bench to the final

boxes, because modules shown at Quick Mode are final; that means a small breakout for the bench. The radio team's driver is written and tested. *(v0.3 had two DP5 modules with patches; v0.2 the Adafruit RFM69HCW at 868 MHz.)*

3. **Board: a custom 2-layer carrier PCB** that the Pico 2 and the radio solder onto. It also carries the radio regulator, the USB-A host port, the battery charger, the 5 V supply for the stick, ESD protection, and the switch, button and LEDs. The antenna is a Molex flex omni stuck on a box wall, its cable plugged into the radio module: no RF circuitry on our board. Before Quick Mode we use a perfboard prototype with the **same modules**.
4. **Power: one protected 18650 Li-ion cell**, charged on board through a charge-only USB-C port. Our whole competition day uses roughly 10–20 % of it.
5. **Estimated cost: about €40–55 per device** with a lab-made PCB, or about €30–40 with an external fab, against the €150 cap.

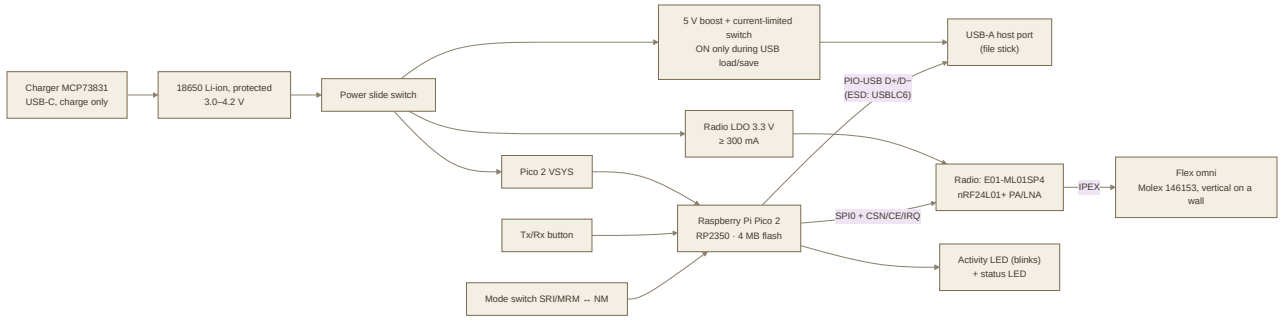
The rules this design answers

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
One and only one module including a microprocessor; no single commercial module embedding the entire system	Pico 2 is the only micro. The radio, charger and boost contain no CPU. No USB-host chips that have their own controller (e.g. CH376), which could count as a second micro.
No chip implementing WiFi, Bluetooth, ZigBee, 2G–5G	**Banned even if unused:** Pico W / Pico 2 W, ESP32 boards, Arduino Nano 33 BLE/IoT, Uno R4 WiFi, Nano ESP32, Raspberry Pi Zero W / 3 / 4 / 5.
Read/write the file from/to a USB memory device: 5 min setup while still wrapped in foil, 3 min to upload afterwards	A USB-A host port on the box wall, plus **≥ 1 MB on-board storage**. The sender copies the file in at setup, and the receiver keeps what it got until upload. The stick can be removed during the round.
Fits entirely in the 15 × 15 × 7 cm IKEA 365+; only connectors, switches and buttons may stick out	PCB about 100 × 80 mm or smaller. The antenna sits inside: a 35 × 9 mm flex omni, vertical, as high as possible on a wall.
Operates autonomously	A battery and a power switch. Nothing plugged in during a round.

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
At least one Tx/Rx switch or button; a pilot light that blinks in Tx/Rx	Panel pushbutton for Tx/Rx and an activity LED. The box is transparent, so the LED can stay inside.
Extra switches allowed (e.g. network mode), but **no switch indicating the position in the network**	One optional SRI/MRM ↔ NM mode switch. Nothing that encodes a node number.
Drop test from table height to concrete (before MRM)	Everything soldered (no sockets or loose headers), board on standoffs, battery strapped, antenna glued down.
ESD and RF exposure compliance	TVS/ESD protection on USB and the buttons. RF power is small (≤ 10 mW e.i.r.p.; ≤ 100 mW e.i.r.p. while hopping in SRI/MRM).
Modules shown at Quick Mode are final	The QM prototype uses the same Pico 2 and the same E01-ML01SP4 as the final board (on a breakout), and the same flex antenna.
Unit cost ≤ €150; lab-made PCB counts €15; everything (screws, foam, wires) goes in the BoM	Cheap modules; consider an external fab for the PCB (see the cost table).

Block diagram

The only fast data path is SPI to the radio, at 31 kB/s (250 kbps). USB file transfer happens only before and after a round.



Micro module options

Ranked against what the board must do: hold a 1 MB file, act as a USB host, use 3.3 V logic, carry no radio chip, stay cheap, and be ready before Quick Mode.

MODULE	STORAGE FOR 1 MB FILE	USB HOST	PRICE	VERDICT	NOTES
Raspberry Pi Pico 2 (RP2350)	4 MB flash ✓	PIO-USB on 2 GPIOs, or native USB	~$5	recommended	Native USB stays free for programming and logs. Big community. Castellated edge, so it solders flat (good for the drop test).
Raspberry Pi Pico (RP2040)	2 MB flash, tight	Same as above	~$4	fallback	Works, but 1 MB of file plus firmware leaves little headroom.
Adafruit Feather RP2040 USB Host	8 MB flash ✓	USB-A on board	~$15	fastest prototype	Less to build, but costs more, and we would show less electronics design of our own (which is graded).
Teensy 4.1	8 MB flash ✓	Host port + library	~$35	expensive	Easiest USB host, but it hurts the cost category.
STM32F411 "Black Pill"	512 KB ✗	Native OTG	~$5	needs extra flash	Would need external flash, and USB mass-storage host

MODULE	STORAGE FOR 1 MB FILE	USB HOST	PRICE	VERDICT	NOTES
					firmware is heavier.
Arduino Uno / Nano (AVR)	✗	✗	–	no	5 V logic (the radio is 3.3 V only), 2 KB RAM, no USB host.
Raspberry Pi Zero (non-W)	SD card ✓	✓	~$5–10	no	Linux: slow boot, high current, and SD corruption on power loss or a drop.
Pico W / Pico 2 W, ESP32, Nano 33 BLE/IoT, Uno R4 WiFi, Pi Zero W, Pi 3/4/5	—	—	–	banned	On-board WiFi/BT chip.
Micro + radio combos (Feather RP2040 RFM69, Moteino)	—	—	–	avoid	Risks the "single commercial module embedding the entire system" rule. Would need an Atenea answer first.

Prices are list prices seen online on 7 Oct 2026 (Raspberry Pi, TME, Adafruit). Get real quotes before ordering.

Suggested parts per device

Example part numbers to start from. Electronics owns the final choice, the footprints and the sourcing.

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Micro module	Raspberry Pi Pico 2 (SC1631), **non-W**	See above	5.5
Radio module	1 × Ebyte E01-ML01SP4 (nRF24L01P + PA/LNA, SMD, IPEX). **Not** E01C-* (clone chip)	Chip-controlled PA/LNA, so only SPI + CSN + CE + IRQ. Same module from the bench (on a breakout) to the final box	≈ 4–6
Antenna	Molex 146153-0100 flex omni (34.9 × 9 mm, 100 mm cable, U.FL)	Plugs straight into the module. Vertical polarisation, mounted high on a wall	≈ 3
USB host port	USB-A receptacle, through-hole, panel-facing	Through-hole survives plugging and the drop test	0.5
5 V for the stick	Boost (e.g. TPS61023) + current-limited switch (e.g. TPS2051 / AP2552), enabled by a GPIO	Sticks need 5 V. Powered only during load/save, which saves energy and protects against shorts	2
ESD	USBLC6-2SC6 on D+/D−; TVS on VBUS and the button lines	ESD compliance; the USB port is touched every round	0.6
Battery	1 × protected 18650 Li-ion (~2500 mAh) with solder tabs, or holder + strap	Steel can is robust in the drop test; rechargeable (sustainability report)	6–8
Charger	MCP73831 (or TP4056) + USB-C receptacle, charge only	Charge between rounds without opening the box. Nothing is plugged in during operation	1.2
Controls	Slide power switch, panel Tx/Rx pushbutton, optional mode switch, 2 LEDs	Required by the rules	2.5

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Radio supply	3.3 V LDO (AP2112K-3.3 or TLV75533, dropout ≤ 0.1 V at 150 mA) + 100 µF / 10 µF / 100 nF at the module; 100 µF on USB VBUS	SP4 VCC is 2.0–3.6 V (abs. max 3.6 V); PA/LNA modules are sensitive to supply noise; handles the ~120 mA TX bursts and stick inrush	1
PCB	2-layer carrier, ≤ 100 × 80 mm	ETSETB lab = €15 in the BoM (rule); an external fab is ~€1 per board but adds 1–2 weeks	1–15
Mechanics	M3 nylon standoffs, base plate, cable ties, hot glue, IKEA 365+ box	Drop test; the BoM must list them too	5–7
Total per device (verify with quotes)			**≈ 30–55**

Power budget

Current draw

Pico 2 running	≈ 30–50 mA ESTIMATE
SP4 TX	≈ 120 mA peak DATASHEET
SP4 RX	≈ 26 mA DATASHEET
USB stick (5 V)	50–100 mA, only during load/save ESTIMATE
Peak on the radio LDO	≈ 120 mA (the radio does not load the Pico's 3V3)

Battery sizing

A sending round averages about 150 mA. Idle with the radio listening is about 70 mA. A full competition day of ~4 hours powered on needs about **300–600 mAh**.

One 2500 mAh 18650 covers that about four times over. A smaller LiPo would also do, but pouch cells are more fragile in the drop test.

Low battery: the LDO is fine

The LDO gives a clean 3.3 V down to a cell of about 3.4 V (dropout ≤ 0.1 V at 150 mA). Below that it passes the cell through: about

For the sustainability report

We can measure watts per round directly on this design. The parts are rechargeable and RoHS. The modules unsolder and can be

2.95 V at the 3.0 V cutoff, still inside the SP4's 2.0–3.6 V. The cell spends ~90 % of its charge above 3.4 V. An LDO after the 5 V boost (~60 % efficient, boost ripple at the radio) or a 5 V battery would be worse.

- **Battery sense:** an ideal diode (e.g. LM66100) instead of the Schottky into VSYS, so ADC3 reads the cell.
- **Firmware:** ≥ 3.5 V normal; 3.4–3.5 V warning LED; below 3.4 V at setup a loud warning (never block a round); below 3.2 V write received data to flash after every burst.
- **T0** measures the radio output at 3.6, 3.3 and 3.0 V.
- **Fallback:** a buck-boost (TPS63001/TPS63020) if T0 shows more than 2 dB less output at 3.0 V; it needs a filter at the module.

reused, and the board shape lets you repair rather than replace.

Build stages

STAGE	BOARD	GOAL	TARGET
1 · Bench	Pico 2 + E01-ML01SP4 on a small breakout (1.27 → 2.54 mm) with its flex antenna, USB powered	Radio tests T0 (output power) and T2; USB-stick read/write via PIO-USB	`as soon as parts arrive`
2 · Quick Mode	Perfboard, **same module part numbers**, external power allowed	One error-free line from a file in < 2 min, first attempt	`week 6–7`
3 · PCB rev A	Custom 2-layer carrier (KiCad), boxed, battery	Autonomous, drop-tested, range tests T4/T5	`week 8–9`
4 · PCB rev B	Fixes from rev A	Pretest (4 Dec) fully packaged	`week 10–11`

Risks and how we cover them

RISK	MITIGATION
PIO-USB host fails with some USB sticks	Test 3–4 different sticks in stage 1. The RP2350's native USB can also be a host, so a second path exists on the same chip.
The foil-wrapped setup makes the USB port hard to reach	Put the port on a side wall. Ask on Atenea whether the foil may be opened at the port during the 5-min setup.
Brown-out or noise during radio TX	Separate radio LDO and bulk capacitance at each module; battery sense on ADC3 with firmware thresholds; check the radio at 3.0 V in T0 and the box at low charge in stage 3.
Clone radio chips (Si24R1)	Buy only the plain E01-ML01SP4 part number from Ebyte's store or a distributor; T0/T2 check the chip.
SMD module needs a breakout before any PCB exists	A one-evening lab board (or careful wiring) in stage 1; the carrier PCB takes over in stage 3.
Drop test breaks a joint or ejects the battery	No sockets; standoffs; strapped cell; drop rev A on purpose in week 9.
The "attacks" interrupt the link mid-round	Watchdog and brown-out reset; firmware resumes from the last acknowledged line (protocols team).

What we need from electronics

1. Agree or counter-propose the micro by **Fri 16 Oct**, so we can order together with the radio modules.
2. Confirm the E01-ML01SP4 and the radio LDO, say who makes its bench breakout, and pick the PCB route (lab at €15, or external fab).
3. Pin map for SPI and the radio's CSN, CE and IRQ (proposal: GP16/18/19, CSN GP17, CE GP20, IRQ GP21), PIO-USB D+/D–, button, LEDs and mode switch. We will write the radio's hardware layer against it.
4. The IKEA box's inside height and where the antenna can sit highest.
5. Book Prof. Chávez or Prof. Jiménez to review this block diagram and the lab's PCB capabilities.

Radio details (interface pins, channel plan, test plan, 20 dBm hopping in SRI/MRM): Team C Radio Brief v0.7.1. Rules quoted from *MTP F26 – Program Competition Rules – Network Upgrade III + Sustainability*.

Sources checked 7 Oct 2026: Pico-PIO-USB (USB host/device via PIO, RP2040 and RP2350) · Pico 2 specs · Ebyte E01-ML01SP4 manual · nRF24L01+ Product Specification.